

CMSE 801 - Day 11



Testing, modifying, and visualizing
compartmental models

October 2, 2024

General Course Announcements

- **Reminder:** Homework 2 is on the course website and is due by **11:59 on Friday, October 11.**
- I'll be working on grading your quizzes this week.

Questions/comments from Day 11 PCA

- Getting all of the derivative functions defined correctly for each model was challenging
- Keeping track of all the different models and their corresponding parameters for the purpose of making all of the various plots took a lot of effort.
- Still some general issues getting solve_ivp working.

The "Seven Step Process" for using `solve_ivp`

1. Make sure you have a clear set of ODEs where each variable has it's own derivative equation
2. Define a function that returns the value of each derivative using equations and the appropriate variables and model parameters.
3. Create a variable that defines/stores your initial conditions
4. Create a variable that contains all of the time values you want to use for evaluating your model
5. Call `solve_ivp` to get the solution, making sure to pass in model parameters as necessary
6. Unpack the solution into separate variables
7. Visualize and interpret the results

Let's take a look at what the final plot could have looked like

I will also share a video from another instructor that walks through the solutions to the second and third zombie models!

Goals for today

- For the model your group is assigned, you'll run the basic compartmental model and then explore the properties of the model and how changes to the model influence the results.
- As a group, you'll want to think about how you can modify or enhance the model, try to implement those ideas, and study the impact of those changes.
- Collectively, you'll create a slide or two that showcases your model and presents your findings
 - We'll try to go around the room at the end of class and each group will present (as time permits)

Looking forward - Day 12 PCA: A reminder/review of computational models and exploring data with the Pandas data analysis library

- Pandas is the most commonly used data analysis package/library in Python and will be the primary tool we use for doing data analysis in the course
- You'll explore some of the basics of pandas
- You'll spend a bit of time reflecting on what we mean when we talk about scientific models and computational models -- we've been working with models already in class, but it's good to pause and make sure you know what we mean when we talk about model!

